

Amirreza Aleyasin

Computational Biologist

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M.Sc. computational biologist with hands-on experience in deep learning for sequence-structure prediction, protein language models, and HPC-scale training pipelines. In my thesis in Max-Planck institute I worked on a deep learning based peptide-MHC binding predictor to generalize over unseen alleles. I'm looking for a PhD position where I can build on this foundation in a computationally rigorous group.

Education

Oct 2022 – Oct 2025	M.Sc. in Computational Biology and Bioinformatics GPA 1.6 Georg-August-Universität Göttingen , Germany
Oct 2018 – Jan 2022	B.Sc. in Cellular and Molecular Biology GPA 1.7 Azad University, Science and Research Branch , Tehran

Research Experience

Jan 2025 – present	Söding Lab , Quantitative & Computational Biology Max Planck Institute for Multidisciplinary Sciences , Göttingen Internship → Master's Thesis → Research Assistant My work covers sequence-structure representation learning using structural alignment (FoldMason) and multiple sequence alignment (MAFFT), deep-learning model training with custom 2D masked self-attention for pairwise positional interaction modeling, transfer learning to underrepresented alleles, protein language models (ESM), and likelihood-based loss design for noisy binding labels. Outputs: PMBind (master's thesis; manuscript in preparation); PMGen (preprint, second author).
Apr 2024 – present	Hauschild Lab , Predictive Deep Learning for Medicine and Healthcare University Medical Center Göttingen / Justus Liebig University Giessen ML Internship → Research Assistant (FAIRPaCT consortium) Multi-modal survival modeling from somatic mutations (MAF), RNA expression, and clinical variables across federated clinical sites. Outputs: FedPanod (PyPI), FRSF4POD (PyPI, GitHub).
Nov 2023 – Jul 2024	Dönitz Lab , Onkopus project University Medical Center Göttingen Student Assistant Contributed to cancer variant interpretation tooling: built a dynamic KEGG pathway visualizer, updated the pyliftover package, and wrote a GitLab auto-merge pipeline used by the team.
Apr – Sep 2023	GippLab , Georg-August-Universität Göttingen Tutor, Python Programming for Data Scientists Supported students during exercises, built web scrapers, and curated an NLP dataset for media-bias research.

Independent Projects

Apr 2026	bayesian-playgrounds : an interactive webpage for teaching Bayesian and simulation-based inference.
Jan 2026	Kaggle: Stanford RNA 3D Folding Part 2: Gold medal . Built a pipeline for RNA 3D structure prediction combining homology modeling, MSA-guided folding, deep learning for coordinate prediction, and PDB structure analysis across diverse RNA families.
Nov 2025	Kaggle: CAFA 6 Protein Function Prediction : GNNs, BioBERT, MMseqs2, ESM-3.

Skills

Languages	English (C1) German (B1) Persian (Native)
Programming	Python (TensorFlow, Keras, PyTorch, scikit-learn, Pandas, NumPy, BioPython, Biotite, RDKit), R, Bash, LaTeX, TypeScript, React
ML / Data Sci.	Deep Learning, Representation Learning, Transfer Learning, Federated Learning, Template-Based Modeling, NLP, Simulation-based Inference (Bayesian), Survival Analysis
Bioinformatics	RNA secondary and 3D structure prediction, template-based modeling (TBM), Kabsch superimposition, sequence-structure alignment (FoldMason, MAFFT), protein language models (ESM), multi-omics integration, genomic variant processing (MAF/VCF), RNA-seq
Infrastructure	Docker, Git/GitLab CI/CD, HPC/SLURM, Conda/Mamba, Linux

Publications

Nov 2025 **PMGen: From Peptide-MHC Prediction to Neoantigen Generation** (preprint).
Second author, benchmarking and validation. DOI: [10.1101/2025.11.14.688404](https://doi.org/10.1101/2025.11.14.688404).

Workshops and Certificates

Apr 2025 **AI and Law workshop** – Erasmus BIP, **Universidad del País Vasco**, Donostia-San Sebastián, Spain.
 Sep 2024 **How to learn from complex data I: simulation-based inference** – CIDBN, Göttingen.
 May 2022 **IELTS Academic** – Overall Band **7.0** (CEFR **C1**). Listening 8.5 | Reading 6.5 | Writing 6.0 | Speaking 7.0.
 Mar – Apr 2021 **Molecular Genetics Internship** – **Azad University** with Mana Lab, Tehran. PCR/DNA extraction, RT-PCR, immunoassay.
 Nov 2021 **Statistics and R** – **HarvardX** (edX).
 Oct – Dec 2020 **Evolutionary Computation and Biological Computation** – Interdisciplinary Schools, Tehran.

Funding and Awards

2025 **Niedersachsenstipendium** – Graduate scholarship, State of Lower Saxony.
 2025 **Erasmus+ BIP** – 2x travel and study grant (Spain and Sweden).
 2024, 2025 **FAIRPaCT Consortium Workshops** – 2x travel funding (Munich and Giessen).

Extracurricular Activities

Archery — DHM Barebow Rookie (German University Championships) | **Game design**

References

Dr. Johannes Söding
Quantitative and Computational Biology,
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Prof. Dr. Anne-Christin Hauschild
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